

ABRA — Findings for the World (running paper)

A living dissemination doc. Maintained in the same pass as the work. It accumulates the publishable findings and keeps a platform-ready draft for each channel (academic, blog/Reddit, Twitter/X thread, YouTube). Numbers here must match the model JSON reports, the white paper, and the site. Last updated 2026-07-24.

The one-line thesis

You cannot predict who wins a Champions VGC game from the two team sheets — even a player-Elo model ties a coin. So we stopped predicting outcomes and built a *decision* stack; every claim ships a proper score, a confidence interval, and an honest baseline.

Headline findings (each measured, with a baseline)

1. **Winner-from-sheets $\approx$ a coin.** ~~Player-Elo held-out log-loss 0.687 vs coin 0.693~~ — withdrawn 2026-09-09: no artifact carries it (`engine/predictability.py` writes nothing; `data/skill-variance.json` does not exist). The artifact-backed nulls are items 4, 7 and 8 below. A cloned-policy rollout win% is *below* a coin — systematically inverted (backs fast offensive teams that lose).
2. **The damage engine is exact.** Within 5% of the Smogon damage calculator on 100% of 31 scenarios (median 0%).
3. **RETRACTED — the live board is predictable only in the sense that counting Pokémon predicts it.** PORY mid-game win% log-loss 0.567 vs coin 0.693, calibrated to ECE 1.6%, clustered CI [0.548, 0.583].
4. **Picking the team doesn't beat a coin (honest null).** CHOMP-EV on ~~1,205~~ **1,102** held-out games (`data/chomp-ev.json:n_test` ; the struck count is in no artifact and is withdrawn 2026-09-09): bring quality ties coin/Elo/usage; robust to dropping forfeits; a belief-weighted variant also ties.
5. ****The meta looks rock-paper-scissors, but it's a hint.**** Playstyle cycle (Trick Room $\rightarrow$ Hyper Offense $\rightarrow$ Sand) on only 13–18 games/leg, CIs cross 50% — suggestive, not settled.
6. **Intuitions flatten under data.** After fixing a classifier bug (Charizard = Sun/Mega-Y), Sun went 15 $\rightarrow$ 1,367 teams and Rain vs Sun is even (51/49, $n=236$); Tailwind vs no-Tailwind is 47% ($n=756$).
7. **Single-label archetypes were the bug; multi-label roles pool the data.** Tagging each team by the 52 functional roles it reveals (`data/pokemon-roles.json:roles` ; the "26" typed here was stale) (not one archetype) lifts the median matchup cell from $n \approx 15$ to **$n=20$** across 1,051 cells (2026-07-25, clean games) — still above the old single-label cells of 11–18, but only just. An earlier $n=7,971$ was retracted in 2.7.0 as an over-tagging artifact. But predicting the winner from preview roles still **ties a coin** (0.694 vs 0.693): roles describe and attribute, they don't predict.
8. ****WAR: a null — which species you bring carries no demonstrated signal. A ridge Adjusted-Plus-Minus (RAPM) model on preview species appeared to beat a coin (0.6875 vs 0.6931), a result** withdrawn 2026-07-25** — that figure was measured on the UNFILTERED store. The clean-store figure this item then carried (~~0.7048 against a coin's 0.6931, accuracy 0.502~~) is withdrawn 2026-09-09: `data/war.json` ($n=3,663$, $\lambda=200$ selected on held-out) reads 0.6936 vs coin 0.6931, accuracy 0.504, verdict *worse than a coin at every regularisation strength tested*. The apparent unfiltered signal was four bot accounts playing one team in 1,446 games. Leaders Basculegion / Kingambit / Sylveon are a descriptive ordering only; effect sizes small, ridge-shrunk.

9. **Roles can be discovered, not declared.** NMF of the team×role matrix at **rank 4** — the rank read from `data/nmf-rank-selection.json:most_reproducible.rank` by `engine/nmf_roles.py` since 2026-09-09, not typed — recovers 4 archetypes, composed by the data and named by a human: A1 physical + priority offense (physical attacker 39%, priority attacker 22%; share 27.1%); A2 bulky support + rain (bulky wall / support 23%, special attacker 15%; share 25.7%); A3 spread offense (spread attacker (both foes) 54%, special attacker 8%; share 25.2%); A4 intimidate + fake out control (debuff (intimidate / drops) 42%, fake out (tempo) 26%; share 22.1%) (`data/nmf-roles.json` , regenerated 2026-09-09, `archetype_recon_error` 0.738, 63,882 team-sides × 52 roles). Each team is a *blend*, never one label. Move-level NMF is coarser (offensive cores dominate; `reconstruction_error_ratio` 0.8348). The six archetypes published here before were at a hand-set rank 6 that scored **-0.107** excess over a shuffled null on the project's own criterion (`engine/nmf_rank.py` , bootstrap factor stability, cf. Brunet et al. 2004); rank 4 scores **+0.0775**. **The caveat that travels with this claim:** the selection artifact is from 2026-07-28 at 6 bootstrap pairs (the defence asks ≥ 50) with no cophenetic correlation, so the rank is *read* rather than *defended* until `python engine/nmf_rank.py 50` is run. Reconstruction error was cited here as the justification and has been withdrawn: `nmf_rank.py` states that it cannot select a rank, because it falls monotonically with rank by construction. "Topic coherence is next" was also removed — the 2026-07-31 defence ruled that *next* is not a justification.

1. **The field really does tech for the metagame — measured.** Rare "tech" move slots carry **+0.0386** more meta-weighted type coverage than a species' standard kit (95% CI [0.0155, 0.0617], excludes zero; positive in 90/148 species). Against Kingambit (Dark/Steel, on 29% of teams) the top tech answers are Incineroar's Close Combat and Blastoise's Aura Sphere — both **4×**. Design note: the store spans 3 days, so the temporal test was impossible and a cross-sectional design was used instead; that limitation ships with the number.
2. **A silent data defect can hide an entire mechanic.** The extractor never parsed mega evolution, so 904 of 906 Charizard-Mega-Y sets had no ability, and the damage engine's dex contained ONE mega forme while computing damage for dozens. Raichu-Mega-Y is **No Guard** — which is the whole reason Zap Cannon (50% accuracy) appears on ~590 Raichu sets. A set that looks like a mistake is often a mechanic you are not modelling.
3. **Some facts are unobservable in principle, and should be labelled as such.** A mega's ability is never announced in a replay log, so it cannot be harvested at any sample size — it has to come from the game's own data files. Conversely, Zoroark's Illusion *can* be proven from logs alone: if the apparent species cannot learn the move and Zoroark can, the disguise is certain. **156 proven disguises** across 395 Zoroark team-sides.

Platform drafts (kept current)

Academic abstract (~150 words)

Competitive Pokémon (VGC doubles) is a two-player, zero-sum, imperfect-information, simultaneous-move game with a non-transitive metagame and an astronomically large team-building space. We show empirically, on thousands of real ladder replays, that predicting a game's winner from the two team sheets is near the coin-flip ceiling — a strong player-rating model does not beat it. We therefore frame the problem as decision support rather than outcome prediction, and evaluate every component with a proper score, a clustered confidence interval, and an honest baseline. Our validated damage engine matches the community ground truth; a small mid-game value network beats a coin and is calibrated; and we report two honest negatives — team-preview bring selection ties a coin, and the "rock-paper-scissors" metagame is suggestive but underpowered. We outline a belief-state, depth-limited search capstone (ReBeL/Metamon-style) as future work.

Reddit / blog (r/VGC, r/MachineLearning) — hook

"I collected 5,000+ real Champions games and tried to build an AI that predicts who wins from team sheets. It can't — it's a coin flip, and so is Elo. Here's what I built instead, and the two things I was sure about that the data proved wrong."

Twitter/X thread (numbered skeleton)

1/ You can't predict a competitive Pokémon game from the team sheets. Even a rating model ties a coin. 2/ So I stopped predicting winners and measured *decisions* — every number with error bars. 3/ Damage math: exact. Mid-game win%: beats a coin, calibrated. Team-picking: ties a coin (I report it). 4/ "Rain beats Sun"? Even (51/49). "Tailwind wins"? 47%. Intuitions flatten under data. 5/ The endgame: an in-battle coach (Stockfish-for-Pokémon). Repo + live site below.

YouTube outline

- Cold open: "I tried to predict Pokémon games with AI. It failed — on purpose."
- The ceiling (why sheets can't call the winner) · the pivot (decisions, not outcomes)
- The town of models (MEDI, GURU, XATU, PORY, KING, CHOMP, ALAKAZAM) on the live site
- The honest nulls (why reporting them is the point) · the road to ALAKAZAM · call to action.

Honest limits (always stated)

Permanent winner-prediction ceiling; meta-structure results are small-sample; rollout policy is the residual GIGO; the capstone's strongest version needs cloud-GPU training.

Related work to cite

Metamon (offline RL + transformers, UT-Austin-RPL); VGC-Bench; PokéAgent Challenge (NeurIPS 2025); Aaron Traylor (competitive-Pokémon framing); ReBeL/CFR/DeepStack/Libratus; Nash-averaging (Balduzzi); blade-chest (Chen & Joachims); HodgeRank (Jiang–Lim–Yao–Ye); sports xG/EPV.

Channels checklist (to publish when ready)

- ☐ arXiv / workshop write-up · ☐ r/VGC + r/MachineLearning · ☐ X/Twitter thread
- ☐ YouTube explainer · ☐ Smogon forums / PokéAgent · ☐ portfolio + live site link